pHathom Technologies Inc.

Carbon dioxide removal prepurchase application

2025

General Application

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

pHathom Technologies Inc.

Company or organization location (we welcome applicants from anywhere in the world)

Halifax, Nova Scotia, Canada

Name(s) of primary point(s) of contact for this application

Mark Bohm, Chief Commercialization Officer

Brief company or organization description <20 words

pHathom is advancing an ocean-based bioenergy carbon capture and storage (BECCS) technology that benefits both ocean and forest health while offering significant revenue generating potential from carbon dioxide removal (CDR) credit sales

1. Public summary of proposed project[1] to Frontier

a. **Description of the CDR approach:** Describe how the proposed technology removes CO_2 from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Our innovative CDR approach

[1] We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

pHathom Technologies is a science-first organization dedicated to commercializing a carbon dioxide removal (CDR) technology that offers low cost and scalability while supporting ocean health and sustainable forestry practices.

Our technology can be applied to any biomass facility that has ready access to seawater (oceans and seas) . This unique approach enables low-cost Bio-energy CCS (BECCS) in coastal locations that do not have ready access to suitable geologic storage for CO_2. pHathom's technology has the potential to rapidly scale into a GT/yr CDR credit generation opportunity.

We are advancing a project to deliver our first CDR credits (510 tons by Q2 2027) at a biomass-to-heat facility in Nova Scotia, Canada, building on the successful piloting of our capture technology in late 2024, and further piloting activities that are planned to validate the techno-economics, safety and stakeholder acceptance of our technology in 2025 and early 2026.

Our technology captures point source CO_2 from biomass energy facilities using a limestone /seawater solution, and converting it into a stable bicarbonate for safe, durable ocean storage thru a process known as Accelerated Weathering of Limestone (AWL).

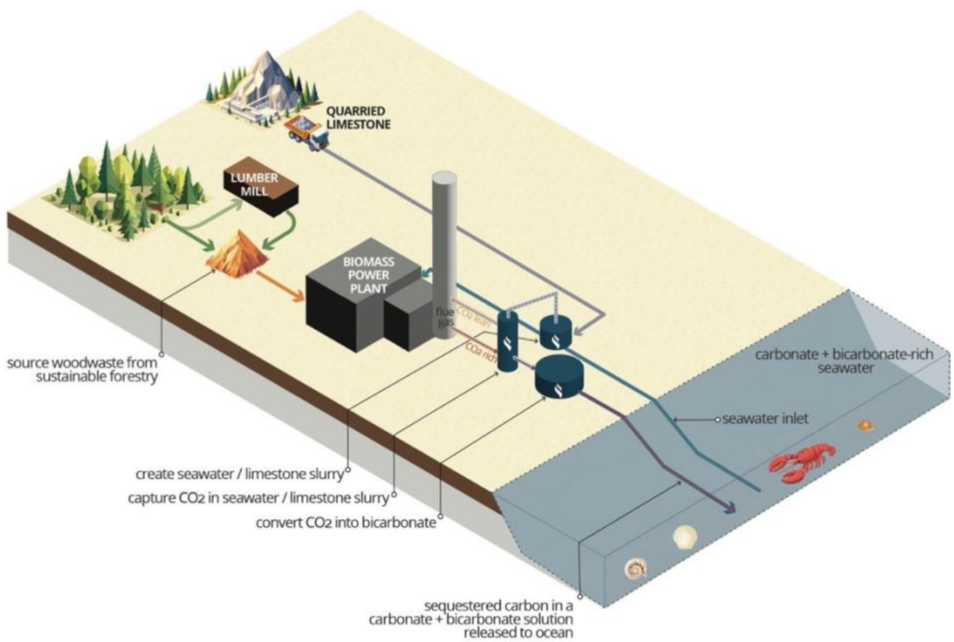

Figure 1: pHathom's CDR process

Accelerating a Natural Process

At its core, pHathom's technology captures CO_2 from biomass power plants in a seawater and limestone slurry which is converted into a stable bicarbonate solution using a process known as Accelerated

Weathering of Limestone (AWL), and stores it in the ocean, where it will be neutral and stable for tens of thousands of years. The process follows the following chemical reactions:

$CO_2 + H_2O \rightarrow H_2CO_3$ (1)
$CaCO_3 + H_2CO_3 \rightarrow Ca^{2+} + 2HCO_3^-$ (2)

Our proprietary process is unique and compelling because we can do AWL in an industrial process configuration, capturing and storing up to 90% of the total facility emissions. This is done by capturing the emissions directly at the stack in a water/limestone slurry contactor, creating bicarbonate-enriched seawater, which is pH adjusted, then returned to the ocean. To accomplish this, we have developed a set of patentable processes (provisional and in progress) that accelerate this process (including CO_2 capture and limestone dissolution rates), reduce water consumption, and increase overall capture rates. In particular, we are developing a proprietary biocatalyst for integration into our AWL process to overcome kinetic bottlenecks in the reactions.

Our technology is best-in-class:

This process is best-in-class as it delivers high quality CDR at a low cost and can scale to a GT/yr removal opportunity; our specific advantages include:

- **Permanent carbon storage:** The bicarbonate solution formed by the process is naturally present and stable in the ocean, ensuring the captured carbon is stored for thousands of years. The ocean holds approximately 50 times more carbon than the atmosphere, and our process simply releases the captured carbon in a stable form that is already naturally abundant in seawater.
- **Scaling potential:** There are over a thousand coastal biomass power plants, large biomass-powered district energy systems and pulp mills globally, and these represent a total capture potential in excess of 350 million tons CO_2/yr. Greenfield plants, as well as fuel switching of fossil to biomass are also an excellent opportunity, with growth in biomass combustion increasing at approximately 6% annually; by 2050 we expect that our technology will be able to achieve gigaton-scale carbon removal; in addition our technology can be applied to emissions from hard to abate sectors, including steel, cement and hydrogen production, furthering its impact on climate.
- **Addressing the negative impacts of ocean acidification :** Our process increases the alkalinity of ocean water. This is significant because ocean acidification, driven by rising atmospheric CO_2 levels, poses a severe threat to marine ecosystems, particularly calcifying organisms such as crustaceans and mollusks. By increasing alkalinity, we help to counteract this acidification, creating a more favorable environment for marine life. The increases in alkalinity and carbonate saturation state that results from our process have the potential to benefit local commercial shellfish populations, such as lobster, clams, and oysters, by mitigating the harmful effects of lowered pH.

- **Support for sustainable biomass production:** pHathom is committed to sustainable biomass sourcing, and to meeting Frontier's Sustainable Biomass Sourcing Principles. We work with partners like Acfor Energy, who ensure that the biomass used in our process comes from sustainably managed forests. This not only minimizes the environmental impact of biomass energy but also creates a market incentive for responsible forestry practices, promoting long-term forest health and biodiversity. We are also working with certification agencies to develop robust tools for reporting and verifying the sustainability of forest management practices.
- **Generation of clean, firm power:** Biomass energy offers a renewable alternative to fossil fuels, but its adoption is currently limited by the higher cost of biomass energy vs conventional fossil sources. pHathom can offer a new revenue steam for biomass operators, which will make biomass energy more economically competitive, promoting its adoption and providing a source of clean, firm power.
- **Cost-effectiveness:** Compared to other carbon capture and storage (CCS) methods, such as those that rely on amine scrubbing and geological storage, pHathom's approach is significantly more cost-effective, and at scale we anticipate coming in below $100/ton. We eliminate the need for energy-intensive solvent regeneration and CO_2 compression, as well as the costly construction of pipelines and injection wells. This makes our technology more economically viable, particularly for smaller-scale biomass power plants.
- **Wider geographic applicability**: Unlike BECCS with geological storage, which is limited by the availability of suitable subsurface geology and the high costs of CO_2 transport, pHathom's ocean-based storage solution can be deployed in any coastal location. This greatly expands the potential for BECCS, making it a viable option for a large number of existing and future biomass power plants around the world.
- **Applicable to small and large sources of** CO_2**:** pHathom's technology can be applied to both small (~15k t/yr CO_2) and large sources of CO_2, whereas BECCS with geologic storage is limited to large-scale sources that can justify the significant investment in CO_2 pipelines and wells. This makes our solution more versatile and able to address a wider range of emission sources.
- **MRV thru direct and continuous measurement** Because our process is onshore, we can make direct total alkalinity, pH, dissolved inorganic carbon and/or PCO_2 measurements upstream and downstream of our process to calculate how much CO_2 is captured, neutralized, converted to bicarbonate and stored. In addition, we will establish a mesh network of deployed sensors in the ocean where the outflow occurs to detect residence time, bicarbonate buildup and potential calcium carbonate precipitation, which would indicate outgassing. Using these long-term, distributed measurement methods, we not only measure how much CO_2 gets converted to bicarbonate and stored in the water, but also gain assurances of its durability and reduce uncertainty by a factor of 10X vs. conventional approaches.

Differentiation from Competitors:
While other companies are working on high quality CDR process, such as BECCS with geologic storage and ocean-based carbon removal, pHathom's approach is unique in several ways:

- Unlike direct ocean capture approaches that extract CO_2 from seawater using energy-intensive electrodialysis (and still require geologic storage), we leverage the concentrated CO_2 stream from biomass combustion.

- Unlike ocean alkalinity enhancement (OAE) approaches that add alkalinity directly to the ocean to absorb atmospheric CO_2, we capture a point-source CO_2 stream, offering more precise measurement and faster carbon removal.

- Unlike conventional BECCS with geologic storage we eliminate expensive CO_2 transportation infrastructure and are not limited to locations with suitable geology.

Our solution helps address the fundamental challenge identified by the IPCC: to achieve net-zero by 2050, we need 250 billion tons of BECCS deployment, but conventional geologic storage cannot scale to this level due to geographic constraints. pHathom's marine storage approach eliminates these constraints while maintaining permanent, verifiable storage.

Alignment with Frontier's Priority Innovation Areas

- **Weathering and mineralization:** Our technology accelerates the natural process of limestone weathering to capture CO_2 and produce stable bicarbonate, which is then stored in the ocean. This approach falls squarely within Frontier's interest in enhanced weathering and mineralization pathways.
- **Biomass-based carbon removal & storage:** Our solution is designed to work in conjunction with biomass power plants, capturing CO_2 from their flue gas. We are also working with sustainable biomass suppliers to ensure that the biomass used in the process is sourced responsibly and is traceable, promoting forest health and avoiding negative environmental impacts, in alignment with Frontier's Sustainable Biomass Sourcing Principles.
- **Marine carbon removal:** Our approach leverages the ocean's natural ability to store vast amounts of carbon in the form of bicarbonate. By storing the captured carbon in the ocean, we tap into a massive and readily available carbon sink. We are working with marine biologists and fisheries to identify and leverage co-benefits of our process to support shell-forming marine life, such as oysters and lobsters.
- **Direct air capture (DAC) integration:** While our primary focus is on capturing CO_2 from biomass power plants, our technology could also be integrated with direct air capture (DAC) systems, offering a novel and scalable carbon storage solution for DAC systems, unlocking the potential for DAC in regions of the world that do not have suitable geology for deep injection of CO_2.

b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?[2] What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

pHathom is currently advancing a pre-commercial demonstration facility in at the Nova Scotia Community College (NSCC) Strait Campus that will serve to generate the CDR credits under consideration for this application. This will be the culmination of three-phased project that will advance the technical, regulatory, stakeholder and economic aspects of the technology, and position pHathom to scale the technology to achieve megaton-scale carbon removal by 2030.

Location and Scale: The project will be executed in three phases, with increasing scale and broader geographic application

Phase 1 & 2 - Pilot Project: Capture and Storage Validation

- **Phase 1 - CO_2 Capture Pilot (NSCC Lunenburg, NS) Q2 2025:** We will conduct a pilot test of our carbon capture system at Acfor's existing biomass boiler at the Nova Scotia Community College (NSCC) campus in Lunenburg in Nova Scotia. This initial phase will allow us to validate the capture process under real-world conditions and provide data for system optimization. The pilot system will have a capture rate of approximately 0.1 tons of CO_2 per day. This phase is crucial for demonstrating the efficacy of our capture technology with real-world biomass flue gas.
- **Phase 2 - Ocean Storage Validation (Huntsman Marine Science Centre, NB) Q3 2025:** We will validate the ocean storage component of our process at the Huntsman Marine Science Centre in New Brunswick. This will involve mesocosm studies and open-water trials to ensure the safe and effective storage of bicarbonate in the ocean. The system will have a capture rate of approximately 0.3 tons of CO_2 per day and will incorporate our proprietary biocatalyst which is expected to accelerate CCS by 5X. This phase is critical for validating efficiency gains as well as demonstrating the environmental safety and long-term stability of our storage approach. It will also identify expected co-benefits to local fisheries and marine life.

Phase 3 - Pre-Commercial Demonstration (NSCC Strait Area Campus, NS) Q2 2027:

- This phase represents our first permanent, ongoing installation, where we will capture 1-4 tons of CO_2 per day, and will generate the CDR credits that will be delivered to Frontier as part of this pre-purchase agreement.

[2] We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at $100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

- It is expected that this facility will continue operation beyond the delivery of the credits to Frontier, in partnership with Acfor Energy and the Nova Scotia Community College (NSCC)

Current cost breakdown: Our current cost structure at the pilot scale is approximately \$980/ton CO_2, broken down as follows:

- Limestone: \$70/tCO_2 (2.5 tons limestone @ \$28/ton delivered)
- Water moving: \$45/tCO_2 (3,300m³ water per ton CO_2 @ \$0.108/kWh)
- Electricity: \$18/tCO_2 (for system operations excluding moving water)
- CAPEX: \$535/tCO_2 (first-of-kind system, sub-optimal size)
- OPEX: \$174/tCO_2 (full time people watching a small system)
- Ocean monitoring and verification: \$138/tCO_2 (initial system over-specified for baseline data)

Achieving Frontier's cost criteria:
To reach Frontier's cost criteria of <\$100/ton and achieve climate-relevant scale (0.5 Gt CDR/year), we are advancing several key improvements:

- **Process optimization and technology deployment:** We are focused on improving the efficiency of our capture and neutralization processes to reduce water and energy consumption. This includes optimizing limestone dissolution rates, reactor design, and water management. It also includes incorporation of biocatalyst to increase the rate of CO_2 capture which directly reduces pumping costs, and the size of the equipment required (and thus capex). We are also working to improve the reaction kinetics of the process, which will allow us to use smaller reactors and reduce residence times (further reducing capex).
- **Economies of scale:** Scaling up our operations will significantly reduce per-unit costs. More facilities will allow us to leverage bulk purchasing of limestone and other materials, as well as optimize labor and operational expenses across facilities. As we move from pilot-scale to commercial-scale deployment, we expect to see substantial cost reductions in both capital and operating expenses.
- **Modular design and manufacturing:** We are developing a modular system design, initially focused on capturing 15 kt/yr that can be easily manufactured and deployed at multiple sites. This will streamline production, reduce capital costs, and enable faster scaling. The use of standardized, modular units will also allow for a more distributed deployment model, enabling us to serve a wider range of biomass power plants.
- **Long-term limestone sourcing:** Establishing long-term contracts for bulk limestone sourcing and exploring opportunities to utilize low-cost waste sources, such as limestone fines, steel slag and mine tailings, which could further lower our costs and reduce the environmental impact of our operations.

Our approach to carbon removal accounting:

We are committed to accurate and transparent measurement, reporting, and verification (MRV) of carbon removal. Our approach involves:

- **Direct measurement**: We will use in-line sensors to directly measure total alkalinity and pH, and periodic sampling to measure dissolved inorganic carbon (DIC) in the water stream, both upstream and downstream of the capture process. This will allow us to directly quantify the amount of CO_2 captured and converted to bicarbonate. The sensors provide continuous, real-time measurements, significantly reducing uncertainty compared to traditional oceanographic methods. Our partner Aquatic Labs' novel sensor systems are the first to measure Total Alkalinity in situ and without reagent, allowing a more accurate calculation of Dissolved Inorganic Carbon (DIC), and will be deployed at our facilities.
- **Ocean monitoring:** We will deploy a network of Aquatic Labs' continouous sensors in the ocean near the discharge location at both Huntsman and the NSCC Strait Area Campus to monitor the long-term fate of the stored carbon. These sensors will track key parameters such as pH, total alkalinity, and dissolved inorganic carbon to verify that the bicarbonate remains stable and to detect any potential impacts on water chemistry. We will also monitor turbidity to detect any potential precipitation of calcium carbonate, which could indicate CO_2 outgassing. This monitoring program will be designed in collaboration with experts in oceanography and marine biology, and will allow a better understanding of how ocean monitoring can be deployed as the technology scales and is deployed at multiple sites.
- **Third-party validation:** We are currently evaluating leading carbon credit standards bodies to certify the carbon dioxide removal credits generated by our process and anticipate initiating our Project Design Document (PDD) by the end of Q3 2025. Collaborating with a recognized standards body will ensure our credits meet rigorous standards for integrity and credibility, providing voluntary carbon market buyers—such as Frontier—with assurance that our carbon removal is real, measurable, verifiable, and of high integrity.

c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

pHathom's BECCS with marine storage approach faces several categories of risks that we have identified and developed mitigation strategies for. We present these risks transparently to demonstrate our commitment to responsible technology development.

CO_2 capture: The basic process relies on reacting dissolved limestone ($CaCO_3$) with flue gas CO_2, which has generally low kinetics and requires significant mass flows of both water and limestone. In addition, adequate mixing and residence time is required to allow for the formation of bicarbonate ions in the solution. While all of this has been successfully demonstrated at lab scale, scaling up the technology to operate at KT to MT/yr range is a risk and is a primary focus area of our IP portfolio and technology development.

Ocean Storage: We are relying on ocean storage for the bicarbonate we create through CO_2 neutralization.

Our biggest risk is not receiving permission from regulators/stakeholders to release our bicarbonate solution to ocean. We are mitigating this risk in multiple ways:

- **Ensuring long-term durable storage:** We are working with marine scientists and building on the existing knowledge around OAE and AWL, along with Aquatic Labs to validate the durable safe storage of the bicarbonate produced in our process.
- By focusing on **modular BECCS (and its smaller scale)**, we minimize the impact of overloading the local seawater with bicarbonate, significantly reducing the risks of losing the stored carbon due to precipitation. With initial deployments in Atlantic Canada, where there is a massive deficit of calcium carbonate concentrations in the seawater (due to CO_2 induced ocean acidification and melting ice and snow from the land and arctic), we **significantly** reduce the early-stage risk, **safely bridging us to understanding the long-term impacts.**
- **Safety/co-benefit Research:** We are working with marine biologist researchers at the Huntsman Marine Science Centre to help identify the levels at which bicarbonate may cause toxicity 1) through chronic exposure and 2) through acute exposure. With this data, we can control our system to ensure that levels never exceed recommended levels.
- **Regulatory Process:** We are working with federal, provincial and local governments to clearly understand the regulatory pathways for gaining permits to release bicarbonate and verify its safety at the levels in which we are working.
- **Community Engagement:** We have begun conversations and will continue to work with local indigenous communities, fisherman, and the broader community to create an open dialogue and build trust. Our solution must be broadly accepted, or it cannot happen.
- We have also partnered with Envigour Consulting, a local specialist in regulatory and community affairs, to ensure that we understand and address stakeholder concerns in a transparent and responsible manner.

Commercialization and Scaleup: Another significant challenge will be developing a commercial-scale integrated package that an end customer (typically a utility operator) will be confident in implementing and operating at an existing or planned biomass plant. Some of the challenges that will need to be overcome include:

- **Development and validation of commercial scale equipment:** We are currently screening potential EPCM partners to help in developing commercial scale equipment that meets the necessary codes, constructability and operability requirements for this type of equipment
- **Integration with existing facilities:** Understanding the required footprint of the additional equipment, along with controls, utilities and other infrastructure will be critical to minimize both capital and operating costs
- **Training of qualified operators and technicians:** It is likely that the skillsets of the operations staff at a typical biomass facility will be inadequate to safely and efficiently operate this type of facility. Developing training and change management tools to effectively equip existing staff, and identifying necessary new staff will be a key part of the commercialization of the technology.

d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) *(should be net volume after taking into account the uncertainty discount proposed in 5c)*	510 tons. This figure represents the net volume of CDR credits we anticipate generating and delivering to Frontier
Delivery window *(at what point should Frontier consider your contract complete? Should match 2f)*	October 2027. This is the date at which we anticipate the contract with Frontier to be complete, with the delivery of the agreed-upon CDR credits.
Levelized cost (\$/ton CO_2) *(This is the cost per ton for the project tonnage described above, and should match 6d)*	\$980/t. This represents the average cost per ton of CO_2 removed over the project's lifetime.
Levelized price (\$/ton CO_2)[3] *(This is the price per ton of your offer to us for the tonnage described above)*	\$980/t

[redacted]

[3] This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).